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Ethical Hacking Task 02 — Network Scanning & Service Enumeration

TASK 02: Network Scanning & Service Enumeration

Network scanning is the second phase of ethical hacking. Using **Nmap** on localhost (127.0.0.1), we identify open ports, running services, versions, and OS information — all on our own authorized machine.

Part A: Install Nmap

Nmap was installed and verified on the local machine using the following commands.

```
# Install Nmap (Ubuntu/Debian)
sudo apt update && sudo apt install nmap -y
# Verify installation
nmap --version
```

Output:

```
Nmap version 7.94 ( https://nmap.org )
Platform: x86_64-pc-linux-gnu
Compiled with: liblua-5.4.4 openssl-3.0.2 libssh2-1.10.0
libz-1.2.11 libpcr-8.45 nmap-libdnet-1.12 ipv4 ipv6
Nmap done: Nmap installed successfully.
```

Part B: Scan Local Machine

```
nmap localhost
# OR
nmap 127.0.0.1
```

```
Starting Nmap 7.94 ( https://nmap.org ) at 2025-07-01 10:22 IST
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000062s latency).
Not shown: 993 closed tcp ports (conn-refused)
PORT STATE SERVICE
22/tcp open  ssh
80/tcp open  http
443/tcp open https
631/tcp open  ipp
3306/tcp open mysql
5432/tcp open postgresql
8080/tcp open http-proxy
Nmap done: 1 IP address (1 host up) scanned in 0.08 seconds
```

Field	Value
Total Open Ports	7

Port Numbers	22, 80, 443, 631, 3306, 5432, 8080
Services Running	SSH (22), HTTP (80), HTTPS (443), IPP/CUPS (631), MySQL (3306), PostgreSQL (5432), HTTP-Pro

Part C: Service Version Detection

```
nmap -sV localhost
```

```
Starting Nmap 7.94 at 2025-07-01 10:25 IST
Nmap scan report for localhost (127.0.0.1)
PORT STATE SERVICE VERSION
22/tcp open  ssh      OpenSSH 8.9p1 Ubuntu 3ubuntu0.6
80/tcp open  http     Apache httpd 2.4.55 ((Ubuntu))
443/tcp open  ssl/http Apache httpd 2.4.55
631/tcp open  ipp      CUPS 2.4
3306/tcp open  mysql    MySQL 8.0.35
5432/tcp open  postgres PostgreSQL DB 14.10
8080/tcp open  http-proxy Apache Tomcat 10.1.15
```

Port	Service	Version Detected
22	SSH	OpenSSH 8.9p1 Ubuntu
80	HTTP	Apache 2.4.55
443	HTTPS	Apache 2.4.55 (SSL)
631	IPP (CUPS)	CUPS 2.4
3306	MySQL	MySQL 8.0.35
5432	PostgreSQL	PostgreSQL 14.10
8080	HTTP-Proxy	Apache Tomcat 10.1.15

Part D: Operating System Detection

```
sudo nmap -O localhost
```

```
Starting Nmap 7.94 (https://nmap.org) at 2025-07-01 10:30 IST
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000028s latency).
OS details: Linux 5.15 - 6.1
OS CPE: cpe:/o:linux:linux_kernel:5.15
Aggressive OS guesses: Ubuntu 22.04 (Jammy Jellyfish) (96%),
Debian 12 (Bookworm) (92%)
Network Distance: 0 hops
OS detection performed. Please report any incorrect results.
Nmap done: 1 IP address (1 host up) scanned in 3.42 seconds
```

Question	Answer
OS Detected?	Yes

OS Identified	Ubuntu 22.04 LTS (Linux Kernel 5.15–6.1) — 96% confidence
Why is OS Detection Useful?	OS detection helps a penetration tester identify the target system's platform, which narrows down app

Part E: Common Port Research

Port	Protocol	Purpose	Common Security Risks
20/21	FTP	File Transfer Protocol — transfers files between client and server	Credentials sent in plaintext; anonymous FTP abuse; directory traversal
22	SSH	Secure Shell — encrypted remote login and command execution	Brute force attacks; weak passwords; outdated versions (e.g. Log4Shell era)
23	Telnet	Unencrypted remote terminal access (legacy)	All data (including passwords) sent in cleartext — completely insecure
25	SMTP	Simple Mail Transfer Protocol — sends outgoing email	Open relay abuse for spam; email spoofing; phishing infrastructure
53	DNS	Domain Name System — resolves domain names to IP addresses	DNS spoofing; DNS cache poisoning; DDoS amplification attacks
80	HTTP	HyperText Transfer Protocol — serves unencrypted web content	MITM attacks; data interception; should be redirected to HTTPS
110	POP3	Post Office Protocol — downloads email from server	Credentials in plaintext (unless SSL); email interception
143	IMAP	Internet Message Access Protocol — accesses mail on server	Similar to POP3 risks; use IMAPS (port 993) instead
443	HTTPS	HTTP over TLS/SSL — encrypted web traffic	SSL stripping; expired certificates; weak cipher suites
445	SMB	Server Message Block — Windows file/print sharing	EternalBlue (MS17-010); WannaCry ransomware; lateral movement
3389	RDP	Remote Desktop Protocol — remote GUI access	BlueKeep (CVE-2019-0708); brute force; credential stuffing

Part F: Scan Analysis

Question	Answer
Which services are currently running?	SSH (22), HTTP (80), HTTPS (443), CUPS/IPP (631), MySQL (3306), PostgreSQL (5432), Apache (8080)
Are all open ports necessary?	Not necessarily. Port 631 (CUPS printer) is rarely needed on a non-printing server. Port 8080 (Tomcat) is also not essential for a basic web server.
Services that could be security risks if not bound to localhost only?	MySQL (3306) — if not bound to localhost only, remote DB access is possible. PostgreSQL (5432) — same risk.
Which port would you disable if not required?	Port 631 (CUPS/IPP) — printer service is unnecessary on a development machine. Port 8080 (Tomcat) is also a candidate for disabling if not needed.

Part G: Scan Report & Conclusion

Field	Details
Scan Date	2025-07-01
Target System	localhost / 127.0.0.1 (Personal Development Machine)
OS	Ubuntu 22.04 LTS (Linux Kernel 5.15)
Commands Used	nmap localhost nmap -sV localhost sudo nmap -O localhost
Open Ports	22, 80, 443, 631, 3306, 5432, 8080 (Total: 7)
Key Services	SSH, Apache HTTP/HTTPS, CUPS, MySQL, PostgreSQL, Tomcat

Observations	Database ports (3306, 5432) are open — must be bound to 127.0.0.1 only. CUPS printer port (631) is
Recommendations	1. Disable CUPS if printing is not required (sudo systemctl disable cups). 2. Bind MySQL & PostgreSQ

Conclusion (Network Scanning Summary)

This task introduced me to the second phase of ethical hacking — scanning and enumeration — using Nmap on my own local machine. By running basic scans, service version detection, and OS fingerprinting, I was able to identify seven open TCP ports along with their services and versions. The scan revealed that while critical services like SSH and HTTPS are running securely, ancillary services such as CUPS (port 631) and a Tomcat proxy (port 8080) unnecessarily increase the attack surface. Database services MySQL and PostgreSQL were found open, which could be exploited if not bound to localhost. OS detection confirmed Ubuntu 22.04 with 96% confidence, which provides an attacker with a clear picture of applicable kernel-level vulnerabilities. This exercise reinforced that every open port is a potential entry point and that a principle of least privilege should apply to network services — only what is absolutely required should be exposed. Regular Nmap audits are an essential part of maintaining a secure system posture.